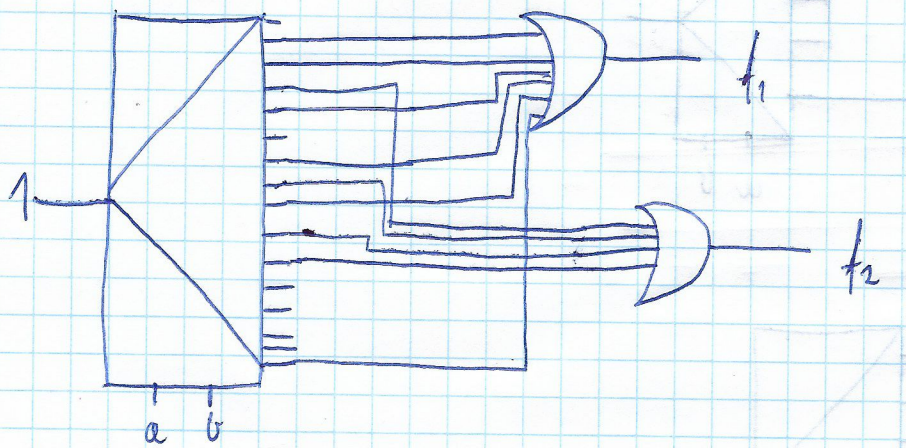
zad 1.

$f_1(a,b,c,d) = \sum m(1,2,4,6,8,15)$

$f_2(a,b,c,d) = \sum m(3,7,9,10) + d(8)$




zad 2

| a b c d | i | f |
|---|---|---|
| 0 0 0 0 | 0 | 1 |
| 0 0 0 1 | 1 | 0 |
| 0 0 1 0 | 2 | 0 |
| 0 0 1 1 | 3 | 0 |
| 0 1 0 0 | 4 | 1 |
| 0 1 0 1 | 5 | 0 |
| 0 1 1 0 | 6 | 0 |
| 0 1 1 1 | 7 | 0 |
| 1 0 0 0 | 8 | 1 |
| 1 0 0 1 | 9 | 0 |
| 1 0 1 0 | 10 | 0 |
| 1 0 1 1 | 11 | 0 |
| 1 1 0 0 | 12 | 0 |
| 1 1 0 1 | 13 | 1 |
| 1 1 1 0 | 14 | 1 |
| 1 1 1 1 | 15 | 1 |

a)

$y_0 = \overline{c+d}$

$y_1 = \overline{c+d}$

$y_2 = \overline{c+d}$

$y_3 = c+d$

b)

$y_0 = \bar{d}$

$y_1 = 0$

$y_2 = \bar{d}$

$y_3 = 0$

$y_4 = \bar{d}$

$y_5 = 0$

$y_6 = d$

$y_7 = 1$

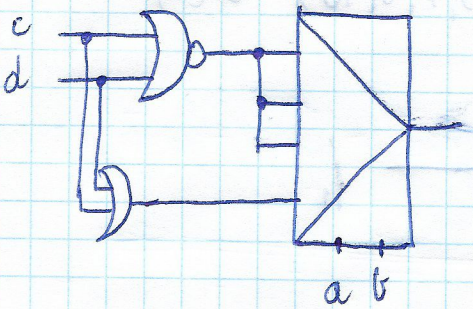
c)

$y_0 = \overline{c+d}$

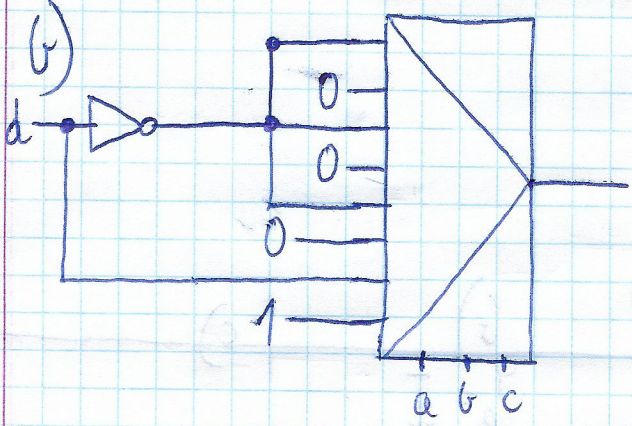
$y_1 = \overline{b \oplus c \cdot d}$

Układy:

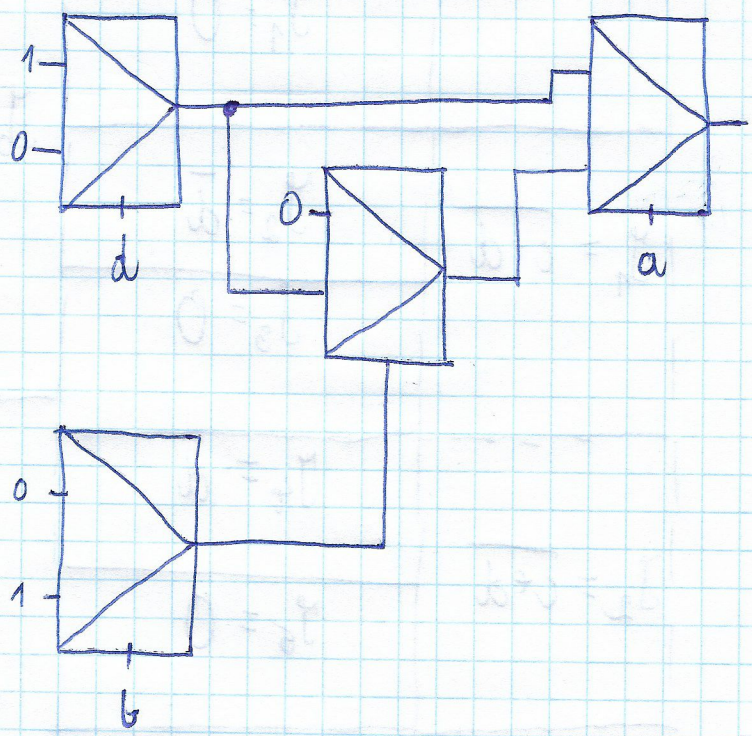
a)




b)




c)

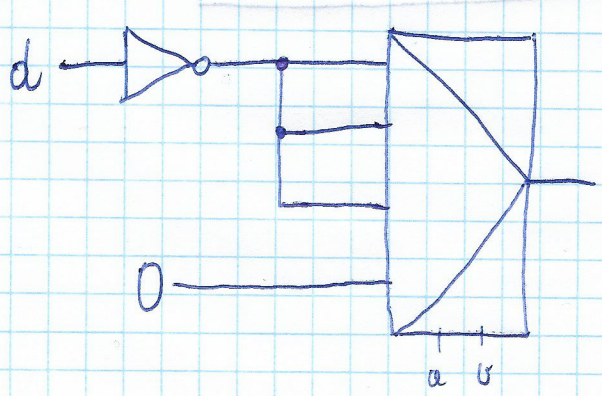


zad. 3

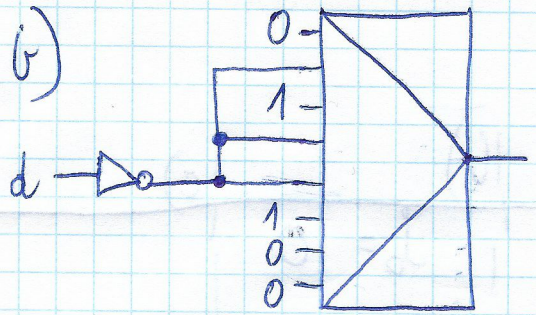
| abcd | i | f | a) | b) | c) |
|---|---|---|---|---|---|
| 0000 | 0 | x | $y_0 = \bar{d}$ | $y_0 = 0$ | $y_0 = \bar{d}$ |
| 0001 | 1 | 0 | | $y_1 = \bar{d}$ | |
| 0010 | 2 | 1 | | | |
| 0011 | 3 | 0 | | | |
| 0100 | 4 | 1 | $y_1 = \bar{d}$ | $y_2 = 1$ | |
| 0101 | 5 | x | | $y_3 = \bar{d}$ | |
| 0110 | 6 | 1 | | | |
| 0111 | 7 | 0 | | | |
| 1000 | 8 | 1 | $y_2 = \bar{d}$ | $y_4 = \bar{d}$ | $y_1 = \bar{b}\,\bar{d}$ |
| 1001 | 9 | 0 | | $y_5 = 1$ | |
| 1010 | 10 | 1 | | | |
| 1011 | 11 | x | | | |
| 1100 | 12 | 0 | $y_3 = 0$ | $y_6 = 0$ | |
| 1101 | 13 | 0 | | $y_7 = 0$ | |
| 1110 | 14 | 0 | | | |
| 1111 | 15 | 0 | | | |

Układy:

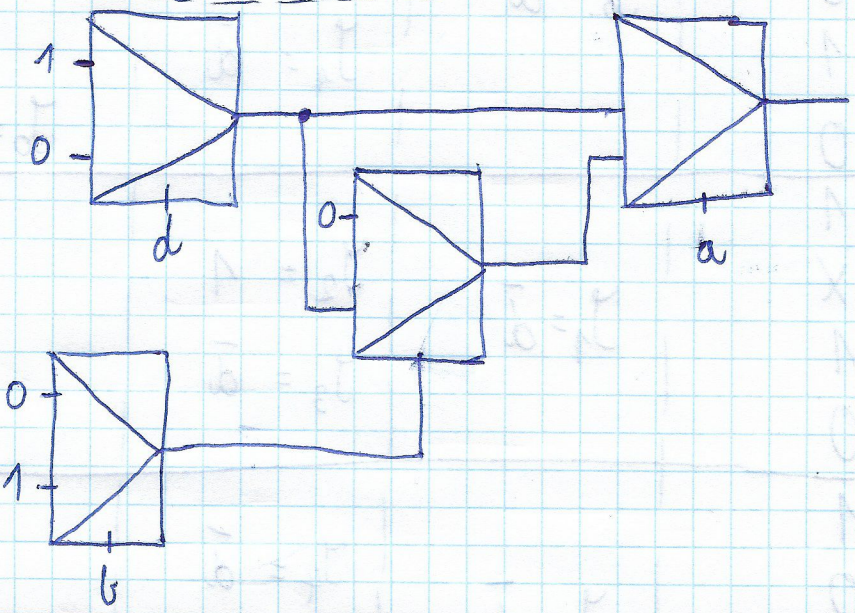
a)



b)




c)







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zad. 4.

$\Sigma(1,2,4,6,7) = \Pi(0,3,5)$

układ:

